

CLASSICAL & MODERN GEOM. OF CIRCLES.

①

PREFACE.

- Want to talk about a perspective on my work which is related to discussions I had with Haha in Leipzig in 2022-23.

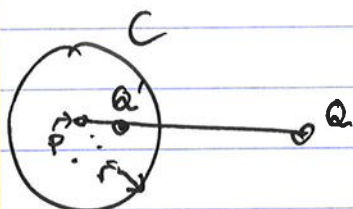
- I learned the classical geometry from Ashold Khovanskii[†] at U. Toronto in 2018.

Also very influential are: ^{*(toric geometry)}

- Coxeter, "Intro. to Geometry"
- Coxeter and Greitzer, "Geometry Revisited"
- Berger "Geometry I & II"
- Sved, "Journey into geometry".

§1. CLASSICAL CIRCLE GEOMETRY.

Defn. (e.g. Steiner, 1820s).

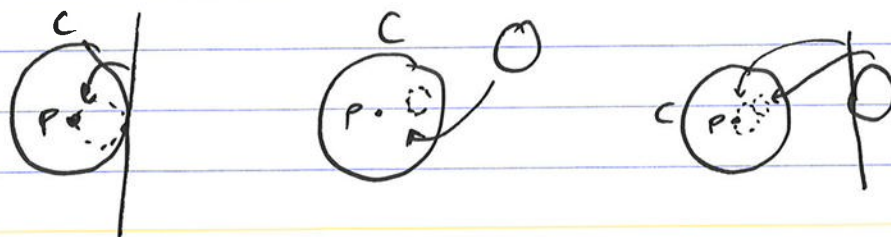


$$|PQ| \cdot |PQ'| = r^2:$$

Q & Q' are inversions of each other in $C = C(P, r)$.

Circle inversion preserves angles & sends circles/lines to circles/lines. (A line is a circle through ∞).

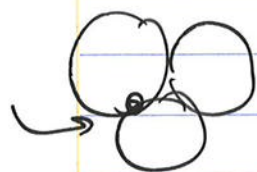
e.g.



(2)

E.g. "Fix any 3 circles tangent in a chain.
There are exactly 2 circles tangent to
all three".

Proof.



invert one tangency point to ∞ .

Get 2 parallel lines & one circle
sandwiched b/w them:



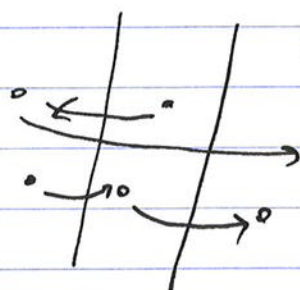
Clearly any circle tangent
to both lines is a
circle of the same rad. as
C that is similarly sandwiched.
Exactly 2 that also meet
C.

□.

Defn. A Möbius transform is a product of an even
number of circle inversions.

Eg. 1.

Translation:



reflect in two parallel
lines.

Eg. 2.

Dilation:



reflect in two concentric
circles.

Eg. 3.

Rotation:



reflect in two intersecting
circles.

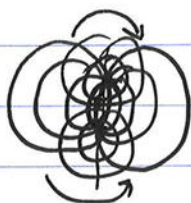
Thm. Every Möbius transformation ($\neq Id$) is one of three kinds:

(i) PARABOLIC.



(= translation, if the single f.p. is at ∞).

(ii) ELLIPTIC:



(iii) LOXODROMIC:



Special case of (iii):
HYPERBOLIC:



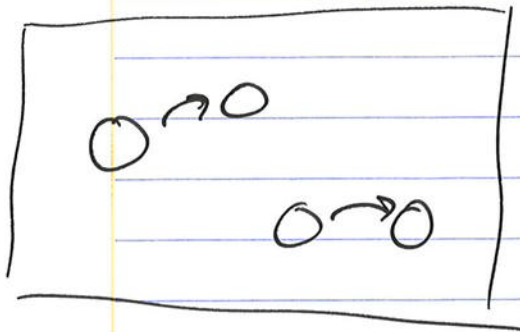
Every LOXO is comp. of a hyperbolic & an elliptic with the same two fixed points.

Thm. Möbius transform = fractional linear maps:

$$\mathbb{C} \ni z \mapsto \frac{az+b}{cz+d} \in \mathbb{C} \quad \text{for } a, b, c, d \in \mathbb{C} \text{ w/ } \begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$$

$$= \underline{\text{PSL}(2, \mathbb{C})}.$$

Eq. 2. Schottky groups (1877).



$2n$ round circles paired up.

Config. space has dimension n

$$G_n - n + n - 6 \rightarrow \text{PSU}(2,1)$$

$\uparrow$ $\uparrow$ $\uparrow$
 $2n$ circles, each with 3 param. (center & radius) can shrink radius of first while enlarge radius of second w/out changing group can choose angle of rotation

$$= G_n - 6 / \mathbb{R}.$$

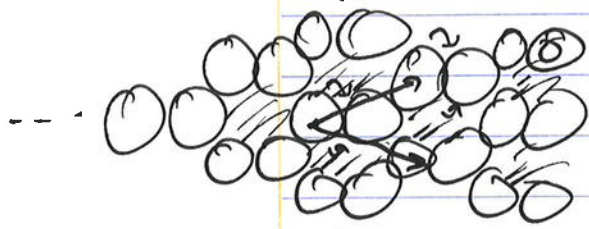
Limit set is a Cantor space (homoeo. to usual middle $1/3$ Cantor set).

Eq. 3.

More complicated: Take a lattice in $\mathbb{R}^2$.

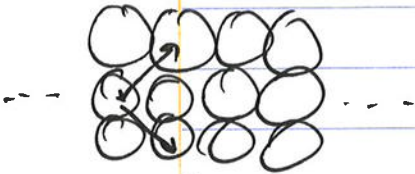
Find a set of tangent pairs of circles which are invariant under the lattice.

3 circles tangent to each circle:



(1 deg. of freedom to stay isosceles)

4 circles tangent to each:



(rigid)

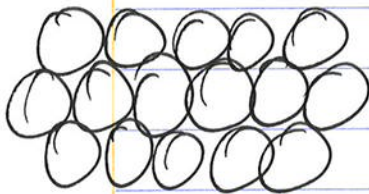
Group gen. by a parabolic pair of adj. circles together with the lattice translations.

5



(rigid)

6



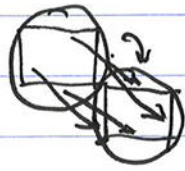
(rigid)

Very intricate circle packings are produced as limit sets.

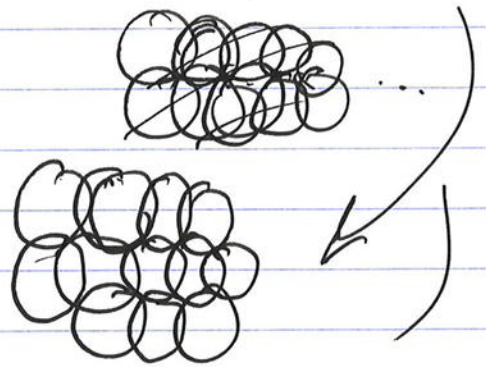
⑦

These figures remind us of one which appeared in
 Krushkal', Aprasov, Gusevskii
 (Akad. Nauk SSSR '81,
 AMS trans. '86)
 from an earlier work of Wieland (1978).

Thm. The group generated by



(so the circle pattern is



is then simplic, as a subgroup of
 $PSL(2, \mathbb{C})$, to

$$\pi_1(S^3 \setminus (\text{circle pattern}))$$

(Note,
 a rigid
 pattern.)
 (Must be,
 by Mostow
 rigidity!)

Not a coincidence!

Thm (e.g. Thurston). For almost any knot
 $K \subset S^3$, there exists a discrete faithful
 repⁿ $\pi_1(S^3 \setminus K) \rightarrow PSL(2, \mathbb{C})$ that arises from
 a pattern of overlapping circles in $\mathbb{R}^2$.

(8)

Major Open Qn. How to compute the circle pattern from R ? How to determine whether a circle pattern arises from a link knot?

§3. CONFIGURATION SPACES.

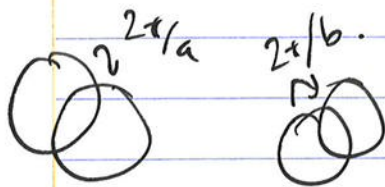
Thm. If a system of circles $\mathcal{D}$ is ^{infinitely} flexible (preserving the initial tangency pattern), then you can push them slightly into each other at points of non-tangency and the resulting group remains discrete & has the same $\mathcal{D}$ -isomorphism class.

(2-lemma of Mariné-Sad-Sullivan; Shokhoshii; Earle-Kra-Krishna')

Qn How far can you push them?

A. In general (unlike classical theory of config. spaces) the resulting space is not semi-algebraic. In fact, it is highly fractal and very wild.

Eg. take two elliptic elements:



The group depends on 1 parameter c

(3x 4 circle params

-2 (angls fixed)

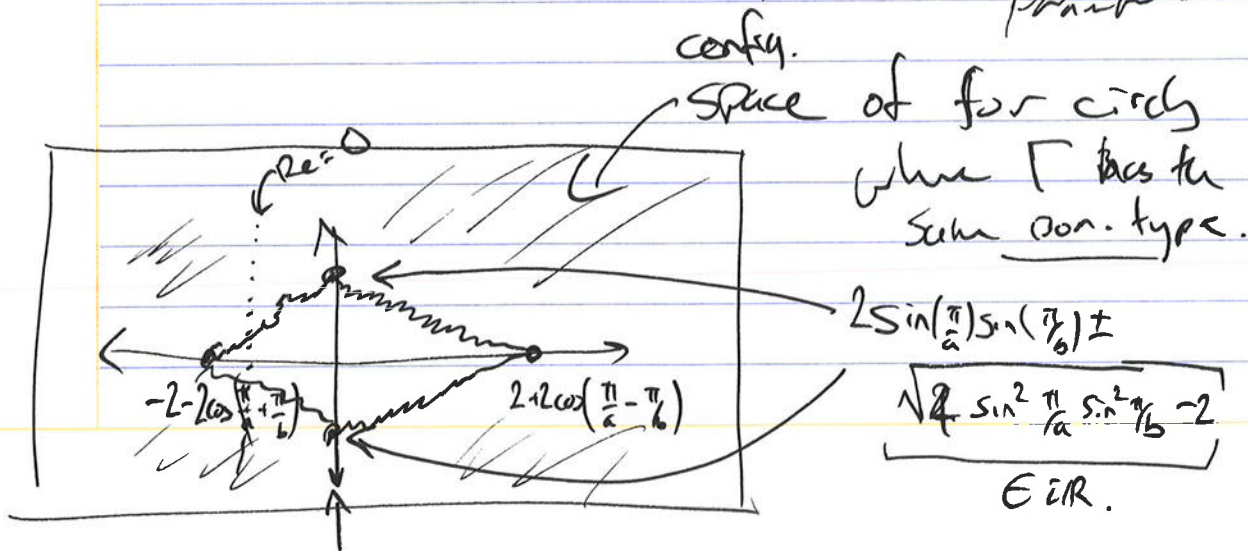
-6 ($PSL(2, \mathbb{C})$)

-2 (pairing elements don't rotate the circles)

$= 2/\mathbb{R}$)

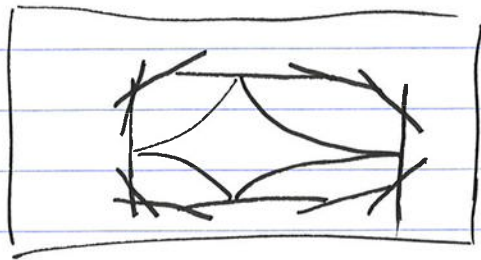
As elts of $PSL(2, \mathbb{C})$:

$$\Gamma = \left\langle \begin{pmatrix} e^{i\pi/a} & 1 \\ 0 & e^{-i\pi/a} \end{pmatrix}, \begin{pmatrix} e^{i\pi/b} & 0 \\ 1 & e^{-i\pi/b} \end{pmatrix} \right\rangle \text{ PGL the single param.}$$



Theorem. (E., Gory, Martin, Schlenker: ^{to appear} AIF '26)

The exterior of the configuration space is contained within an explicit dodecagon, sharp at 4 points!



Application: A conjecture of Ouseley-Morier-Geraud - Veselov:

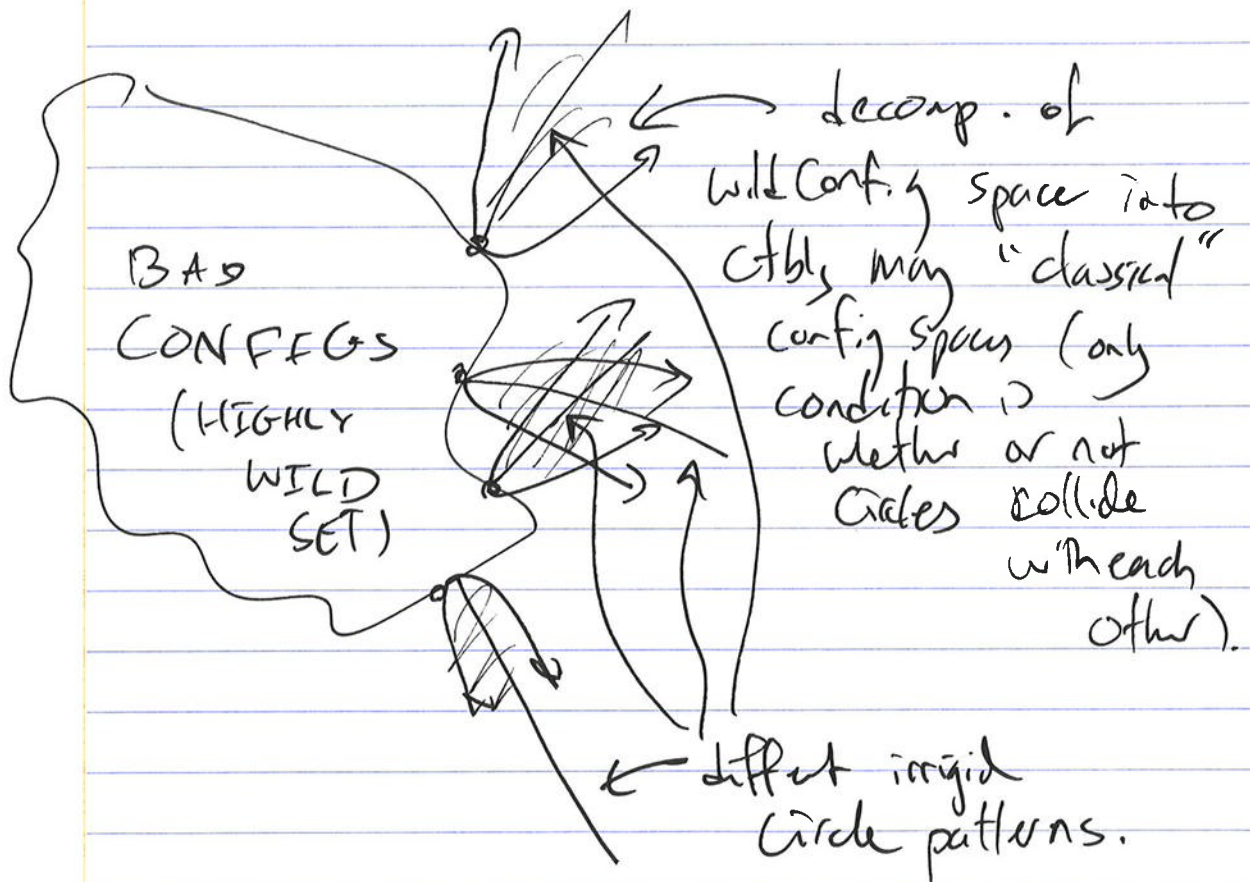
* the power series corresponding to q -rational numbers have radius of convergence $\geq (3-\sqrt{5})/2$.

Equivalently, the Burau rep. $B_3 \rightarrow \text{PSL}(2, \mathbb{C}[[t]])$ is faithful inside the annulus

$$3-2\sqrt{2} \leq |t| \leq 3+2\sqrt{2}.$$

Theorem. (E, arxiv 25).

In arbitrary config spaces of finitely many circles, there is a dense set of points on the boundary such that you can find semi-algebraic ~~sets~~ conditions certifying membership in the space, one sharp at each distinguished boundary part.



For each set you get an rigid circle pattern. Generically, $\exists$ a circle pattern with the "same" combinatorics that gives you a knot group whose geometry reflects the ~~global~~ given subregion of config. space.